

BRANCH DOCUMENT — IRIS-3

Branch: branch_20260512_004625 | Base: main-v1.0

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Decision: NO_GO | Confidence: 43%

CHANGE REQUEST

allunga la struttura rettangolare di base facendola piu lunga ma mantenendo lo stesso materiale

CASCADE ISSUES IDENTIFIED

[CRITICAL] ISSUE-001 — Increased structural length may exceed vibration test limits

Subsystem: mechanical | Req: L1-MEC-002 | Type: constraint_violation

The change in dimensions could alter the modal frequencies and natural modes of vibration, potentially causing the structure to fail the random vibration test at 14.1 Grms.

[MAJOR] ISSUE-002 — Thermal gradient margin may decrease

Subsystem: thermal | Req: L1-THM-001 | Type: constraint_violation

Increasing the length of the base structure can increase thermal gradients across the optical bench, potentially violating the requirement for a maximum ΔT of 15°C during orbital cycles.

[MINOR] ISSUE-003 — Total mass may exceed accommodation envelope

Subsystem: structure | Req: L1-MAS-001 | Type: constraint_violation

The change in dimensions could increase the overall mass of the payload, potentially exceeding the 35 kg limit.

[MINOR] ISSUE-004 — Shock test may need re-evaluation

Subsystem: mechanical | Req: L1-MEC-003 | Type: reanalysis_required

The change in dimensions could affect the shock response characteristics, necessitating a reanalysis of the structure's ability to withstand launch shock loads.

MISSING ECSS TESTS — NOW REQUIRED

ECSS-E-ST-32C Rev.1 §9 — Shock test per launcher SRS ICD

Reason: The change in dimensions may affect the shock response characteristics, necessitating a re-evaluation of the structure's ability to withstand launch shock loads.

ECSS-E-ST-31C §5.4 — Thermal balance test to validate TMM

Reason: The change in dimensions may alter thermal gradients and heat dissipation, requiring a new thermal balance test.

SUBSYSTEM STATUS — UPDATED

Subsystem	Score	Status	Detail	Changed
optical	92%	OK	MTF 0.42 > threshold 0.40	—
thermal	41%	WARN	Thermal gradient margin may decrease	■ MODIFIED
mechanical	53%	FAIL	Increased structural length may exceed vibration test limits	■ MODIFIED
structure	83%	OK	Total mass may exceed accommodation envelope	■ MODIFIED
solar	89%	OK	Power margin +18%	—
startracker	67%	WARN	Stray light analysis pending	—
dataproc	71%	WARN	Compression ratio TBC	—

AFFECTED REQUIREMENTS

- L1-MEC-002 — verification evidence invalidated, re-analysis required
- L1-THM-001 — verification evidence invalidated, re-analysis required
- L1-MAS-001 — verification evidence invalidated, re-analysis required